Unit: 6 Unit 2: Exponents Radicals
Topic: Exponential Equations

Name: ______________
Date: __________

Foundation (Jalapeno)

Q1. Solve for x: $2^x = 8$
(A) $x = 1$
(B) $x = 2$
(C) $x = 3$
(D) $x = 4$
(E) $x = 5$

Q2. Solve for y: $3^y = 27$
(A) $y = 1$
(B) $y = 2$
(C) $y = 3$
(D) $y = 4$
(E) $y = 5$

Q3. Solve for z: $4^z = 16$
(A) $z = 1$
(B) $z = 2$
(C) $z = 3$
(D) $z = 4$
(E) $z = 5$

Q4. Solve for x: $x^2 = 25$
(A) $x = 3$
(B) $x = 4$
(C) $x = 5$
(D) $x = 6$
(E) $x = pm5$

Q5. Solve for y: $y^3 = 64$
(A) $y = 2$
(B) $y = 3$
(C) $y = 4$
(D) $y = 5$
(E) $y = 6$

Q6. Solve for x: $2^x = 32$
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$

Q7. Solve for z: $5^z = 125$
(A) $z = 1$
(B) $z = 2$
(C) $z = 3$
(D) $z = 4$
(E) $z = 5$

Q8. Solve for x: $x^4 = 81$
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = pm3$

Open Ended Questions (FRQ)

Q9. Solve for x: $3^{2x} = 81$

Q10. Solve for y: $2^{y+1} = 64$

Chef's Tips

- Check for correct notation and formatting
- Ensure students show all work and reasoning
- Be available to answer questions and provide feedback